2.3 Time required from co mmencement of cross-flooding 0 until any angle of heel is achieved:

T T Tf =

2.4 Dimensionless factor of reduction of speed through an equaliza tion device, being a function of bends, valves, etc., in the cross-flooding system:

=
k
F 1

where F is not to be taken as more than 1.

Values for k can be obtained from appendix 2 or other appropriate sources.

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SOURCE: MSC.245(83).pdf
function of bends, valves, etc., in the cross-flooding system:

=
k
F 1

where F is not to be taken as more than 1.

Values for k can be obtained from appendix 2 or other appropriate sources.

2.5 Cross-flooding through successive devi ces of different cross-section:

If the same flow crosses successive flooding devices of cross-section S 1, S 2, S 3 having corresponding friction coefficients k1, k2, k3, then the total k coefficient referred to S1 is:

$$k = k1 + k2 \cdot S12 / S22 + k3 \cdot S12 / S32$$

2.6 If different flooding devices are not cro ssed by the same volume, each k coefficient should be multiplied by the square of the ratio of the volume crossing the device and the volume crossing the reference section (which will be used for the time calculation):

$$k = k1 + k2 \cdot S12 / S22 \cdot W22 / W12 + k3 \cdot S12 / S32 \cdot W32 / W12$$

2.7 For cross-flooding through devices in parallel that lead to the sa me space, equalisation

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2.7 For cross-flooding through devices in parallel that lead to the sa me space, equalisation time should be calculated assuming that:

$S \cdot F = S1 \cdot F1 + S2 \cdot F2 +$

with $F = 1/k$ for each device of cross-section Si

3 Air pipe venting criteria

3.1 In arrangements where the total air pipe sectional area is 10% or more of the cross-flooding sectional area, the re strictive effect of any air back pressure may be neglected in the cross-flooding calculations. The air pipe sectional area shoul d be taken as the minimum or the net sectional area of any automatic closing devices, if that is less.

3.2 In arrangements where the total air pipe sectional area is less than 10% of the cross-flooding sectional area, the re strictive effect of air back pr essure should be considered in the cross-flooding calculations. The following method may be used for this purpose:

The k coefficient used in the calculation of cros s-flooding time should take into account the drop

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the cross-flooding calculations. The following method may be used for this purpose:

The k coefficient used in the calculation of cros s-flooding time should take into account the drop of head in the air pipe. This can be done using an equivalent coefficient ke, which is calculated according to the following formula:

$ke = kw + ka \cdot (a/w) \cdot (Sw/Sa)^2$

where:
kw = k coefficient for the cross-flooding arrangement (water)
ka = k coefficient for the air pipe
a = air density
w = water density
Sw = cross-section area of th e cross-flooding device (water)
Sa = cross-section of air pipe

4 Alternatives

As an alternative to the provisions in sections 2 and 3, and for arrangements other than those shown in appendix 2, direct cal culation using computational fluid dynamics (CFD), time-domain simulations or model testing may also be used.

APPENDIX 1

EXAMPLE FOR TREATMENT OF HEEL ANGLES AND WATER HEADS
AT DIFFERENT STAGES OF CROSS-FLOODING

Figure 1

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APPENDIX 1

EXAMPLE FOR TREATMENT OF HEEL ANGLES AND WATER HEADS
AT DIFFERENT STAGES OF CROSS-FLOODING

Figure 1

RESOLUTION MSC.245(83)
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ANNEX 16

APPENDIX 2

FRICTION COEFFICIENTS IN CROSS-FLOODING ARRANGEMENT

RESOLUTION MSC.245(83)
(adopted on 12 October 2007)
RECOMMENDATION ON A STANDARD METHOD FOR
EVALUATING CROSS-FLOODING ARRANGEMENTS
FIGURE 1 FIG!;RE 3
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FICURE 11
FIG URE 9

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